SAR Drone–Rover System

Research and Project Justification

Prepared by: Landon Jin

Spinther Robotics

October 2026

Concept: a lightweight, portable multirotor drone system supported by a forward-deployed mobile charging rover. The rover travels with the rescue operation to reduce repeated drone transit time and sustain aerial reconnaissance. This preliminary report evaluates why such a system could be valuable; it does not prescribe final design specifications.

Executive summary

Small multirotor drones can complement—not replace—ground rescuers and crewed aircraft. Their strongest advantages are quick deployment, hovering over points of interest, viewing inaccessible terrain, and reducing personnel exposure. Published simulations suggest meaningful time savings, while real disaster deployments demonstrate reconnaissance utility. Mortality reductions and financial returns must be validated rather than assumed.

1 | Scale of the problem

Indicator	Figure	Interpretation
NPS SAR people involved (1992–2007)	78,488 total; ~4,906/year	Historical nationwide NPS cases; not all US SAR
NPS SAR fatalities (1992–2007)	2,659 total; ~166/year	Not all preventable by faster detection
NPS classified saves (1992–2007)	13,212 total; ~826/year	Agency outcome category, not all rescues
US billion-dollar weather disasters (2024)	27 events; at least 568 deaths	Broader disaster exposure; not all deaths SAR-addressable
Hurricane Helene (2024)	At least 219 deaths	Illustrates severe access and infrastructure disruption

Source notes: Historical NPS totals are from the 1992–2007 NPS SAR analysis cited in prior research; 2024 disaster figures are NOAA billion-dollar disaster estimates. Confirm the final primary-source editions before external publication.

2 | Economic value of mortality-risk reduction

Using an illustrative value of statistical life (VSL) of $14.2 million, the historical 166 NPS fatalities/year represent 166 × $14.2M = $2.3572B in annual statistical mortality value. This is NOT money a drone would save: it is an upper-bound reference value associated with all recorded fatalities, many of which are not drone-preventable.

Hypothetical reduction in annual NPS fatalities	Expected deaths prevented / year	Statistical benefit / year

1%	1.66	$23.6M
5%	8.30	$117.9M
10%	16.60	$235.7M
20%	33.20	$471.4M

More defensible model: annual societal benefit = (fatalities in drone-eligible incidents) × (measured absolute reduction in fatality probability) × VSL + verified operational savings − full program costs. Treat all values above as scenarios, not forecasts.

3 | Evidence for faster search

Reported finding	Context / caveat
57.3 min → 8.9 min median victim-access time (84.5% reduction)	Mountain rescue simulation comparing ground-only with drone-assisted snowmobile search; 20 simulated operations. Not a universal result.
32,980 vs 1,490 m^2/min (~22.1× area coverage)	Same mountain-rescue simulation; area coverage is not equivalent to successful detection.
3 min 11 sec faster successful searches	EENA/DJI/Black Channel simulated search-team study; ground teams found 17/20 targets and drone teams 23/30, so detection proportion did not improve in that sample.

Interpretation: Faster initial detection and more sustained aerial coverage could help teams prioritize ground resources and shorten the search phase of an incident. These studies show potential, but do not establish a universal reduction in mission duration.

4 | Natural-disaster use cases

Scenario	Multirotor advantage	Measure of success
Floods and hurricanes	Hover over roofs, inspect blocked roads and isolated structures	Time to locate stranded person; mapped sites/hour
Earthquakes / landslides	Inspect unstable structures and slopes without immediate ground entry	Hazard sites mapped; personnel exposure avoided
Wildfires	Reconnoiter evacuation corridors and localized hazards when safe/legal	Reconnaissance latency; actionable coordinates delivered
Remote wilderness	Launch from a backpack or vehicle; inspect ravines and tree lines	Probability of detection; time-to-location

Following Hurricane Helene, the NPS reported 76 drone flights at 56 locations during the first nine operational days on the Blue Ridge Parkway, documenting 21 landslides in approximately 10.6 cumulative flight hours. Drones can help locate stranded or injured survivors, large hazards, and safe passages for disaster relief volunteers to establish themselves quickly after a disaster.

Potential Advantages Over Conventional SAR Drones

A mobile charging rover could improve the operational efficiency of search and rescue (SAR) drones by reducing the time spent traveling between a stationary charging base and the reconnaissance area. By positioning a mobile charging station closer to the active search region, drones could dedicate a greater proportion of their available flight time to reconnaissance rather than transportation.

To illustrate this advantage, consider a hypothetical four-hour SAR operation comparing a conventional drone operating from a stationary base with a drone supported by a mobile charging rover.

Metric	Stationary base	Mobile charging rover
Drone flight endurance	30 min	30 min
Distance to search region	3 km	0.3 km
Flight speed	12 m/s	12 m/s
Round-trip travel time	8.3 min	0.8 min
Productive search time per sortie	21.7 min	29.2 min
Recharge/turnaround time	5 min	5 min

Estimated improvement: Under these assumptions, the mobile charging rover increases productive search time per sortie by approximately 35%, from 21.7 to 29.2 minutes—an additional 7.5 minutes of reconnaissance per flight.

The benefit becomes particularly significant during prolonged SAR operations, where repeated travel between the search region and a stationary charging base consumes valuable flight time. A mobile charging rover could reposition itself closer to the active search area as the operation progresses.

Deploying multiple lightweight drones from one mobile charging platform could also allow alternating reconnaissance and charging cycles, potentially supporting continuous aerial surveillance if charging and scheduling capacity are sufficient.

Engineering significance: Reducing nonproductive flight time could increase search coverage, improve operational endurance, and potentially reduce the time needed to locate missing persons. This may be especially valuable during wilderness searches and natural disasters.

Model limitations: These results are hypothetical, not experimentally validated. Both systems are assumed to have the same endurance, speed, and five-minute turnaround. The calculation excludes safety reserves, takeoff/landing overhead, rover energy consumption, and the time needed to deploy or reposition the rover. The four-hour duration is contextual; the table compares individual sorties, not total four-hour coverage.

Why Search Duration Matters: Survival and Mission Efficiency

Search duration is closely connected to the urgency of locating missing people. In a retrospective study of Oregon search-and-rescue missions from 1997 to 2003, investigators examined 2,302 eligible searches involving 3,287 people. A total of 185 people died (5.6% overall). Search duration was the only variable retained in the final survival-prediction model, with a notable split at 51 hours. The authors reported that searches continuing beyond 51 hours still found survivors in 56.6% of cases; approximately 43.4% in that longer-duration group did not survive. These findings make the

case for timely detection, while emphasizing that 51 hours is a research cutoff—not a typical mission length or a point at which searches should end.

Importantly, the study was observational. Difficult terrain, delayed reporting, injury severity, and other circumstances can contribute to both longer searches and higher mortality. It therefore cannot establish that making an individual search 50% faster will prevent a corresponding percentage of deaths.

Illustrative Opportunity to Shorten Extended Searches

Aerial reconnaissance can inspect difficult or inaccessible terrain while a mobile charging rover reduces repeated return flights to a distant base. The combined system could improve the rate at which a defined search area is covered, especially when teams need to conduct several sorties. If effective coverage rate increases without sacrificing detection probability, the time required to inspect a fixed area may decrease.

Hypothetical effective search rate	Time to search same area (51 h baseline)	Reduction in time
Current rate (1×)	51.0 h	0%
1.35×	37.8 h	25.9%
2×	25.5 h	50.0%
3×	17.0 h	66.7%
5×	10.2 h	80.0%

Calculation: time to search a fixed area = baseline search time / relative effective coverage rate. A 100% increase in search rate means the rate doubles and the search time falls by 50%; time itself cannot decrease by more than 100%. These calculations are illustrative scenarios, not observed improvements from the proposed system. The earlier 35% per-sortie advantage is likewise a theoretical comparison against a stationary base, not a measured 35% improvement in full-mission search speed.

The case for development is therefore the potential to shorten time-to-detection and sustain aerial search coverage across long missions—not a claim that all 51-hour searches can be completed in half the time. Performance must ultimately be measured using comparable search areas, detection rates, and operational conditions.

Sources and verification

- Adams, A. L., Schmidt, T. A., Newgard, C. D., Federiuk, C. S., Christie, M., Scorvo, S., & DeFreest, M. (2007). Search is a time-critical event: When search and rescue missions may become futile. Wilderness & Environmental Medicine, 18(2), 95–101. https://doi.org/10.1580/06-WEME-OR-035R1.1
- Eyerman, J., Crispino, G., Zamarro, A., & Durscher, R. (2018). Drone efficacy study (DES): Evaluating the impact of drones for locating lost persons in search and rescue events. DJI & European Emergency Number Association. https://eena.org/wp-content/uploads/2019/08/2018_09_18_EENA_DJI2Report19.pdf
- Karaca, Y., Cicek, M., Tatli, O., Sahin, A., Pasli, S., Beser, M. F., & Turedi, S. (2018). The potential use of unmanned aircraft systems (drones) in mountain search and rescue operations. The American Journal of Emergency Medicine, 36(4), 583–588. https://doi.org/10.1016/j.ajem.2017.09.025

- National Centers for Environmental Information. (2025). Assessing the U.S. climate in 2024. National Oceanic and Atmospheric Administration. https://www.ncei.noaa.gov/news/national-climate-202413
- National Park Service. (n.d.). Uncrewed aerial systems (sUAS) module supports Hurricane Helene response on Blue Ridge Parkway. U.S. Department of the Interior. https://www.nps.gov/articles/000/uas-module-supports-blri-2024.htm
- U.S. Department of Transportation. (2026). Departmental guidance on valuation of a statistical life in economic analysis. https://www.transportation.gov/office-policy/transportation-policy/revised-departmental-guidance-on-valuation-of-a-statistical-life-in-economic-analysis